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Get started with FinOps

20 min • Module9 Units

- Beginner Administrator AI Engineer Business Analyst Business Owner
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- Technology Manager Startup Founder Service Adoption Specialist Azure
- Dynamics 365 Microsoft Fabric Microsoft 365 Microsoft Defender
- Microsoft Copilot Office 365 SQL Server Microsoft Power Platform
- Industry Solutions

Begin your journey into FinOps on Azure with this module, designed to provide a comprehensive introduction to the FinOps framework. In this module, you'll delve into the essence of FinOps, understanding its fundamental principles, and gaining insights into its core components and best practices.

Learning objectives

By the end of this module, you'll be able to:

- Describe the impact of the shift from a CapEx to an OpEx model.
- Describe the FinOps framework including its principles, domains, capabilities, and more.
- Describe why culture and organizational alignment are important to manage the cloud.
- Identify FinOps stakeholders and understand their motivations.

Attribution - The content of this learn module is based on the [FinOps Framework](#) published by the [FinOps Foundation](#).

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Prerequisites

- Basic understanding of cloud computing concepts

- Basic understanding of pay-per-use model

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Introduction

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Module assessment

Assess your understanding of this module. Sign in and answer all questions correctly to earn a pass designation on your profile.

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Introduction

3 minutes

The shift from CapEx to OpEx model

Running business workloads on cloud platforms enables organizations to enhance reliability and performance while optimizing the cost of computing, storage, and other services. Currently, Software as a Service (SaaS) cloud offerings such as Microsoft 365 present the largest segment in end-user spending, but cloud infrastructure as a service (IaaS) and platform as a service (PaaS) are rapidly growing.

For organizations transitioning from on-premises to cloud environments or managing current workloads in the cloud, shifting from a traditional Capital Expenditure (CapEx) model to a more flexible Operational Expenditure (OpEx) model signifies a major transformation. This change has substantial effects on both the financial and operational sides of the business.

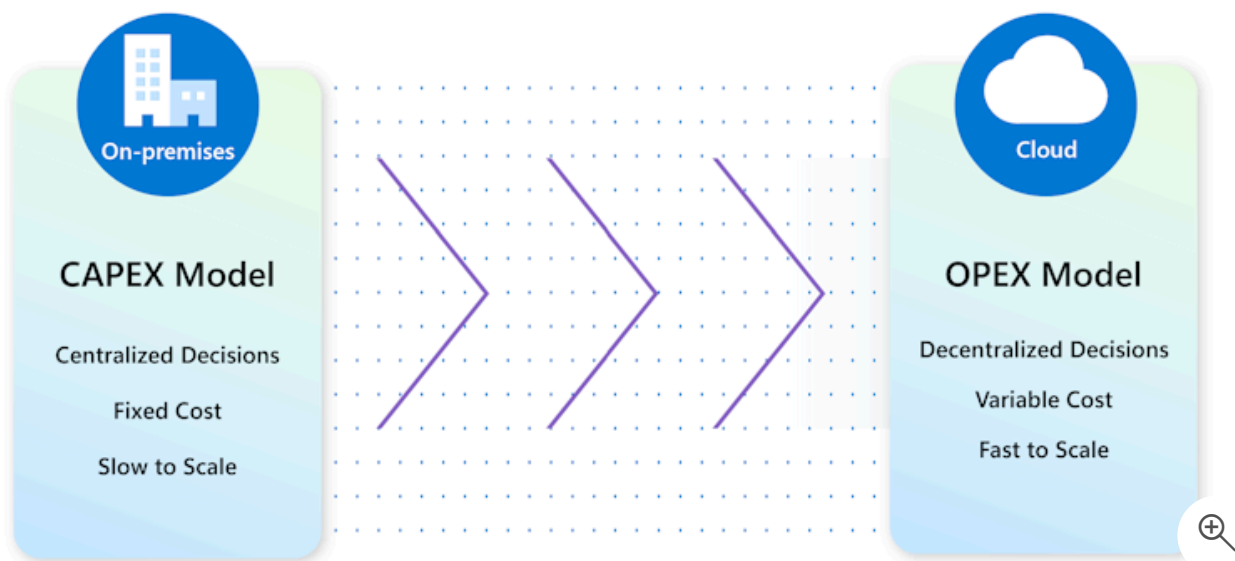
In an on-premises environment, companies typically invest significant upfront capital in physical hardware, data centers, and infrastructure. These assets represent a substantial one-time expenditure that's usually depreciated over several years. Organizations bear the costs of purchasing, maintaining, and upgrading their hardware, which often results in high initial expenses. Moreover, making changes or scaling resources in on-premises environments can be a time-consuming process. This slow time to make changes can hinder an organization's ability to respond quickly to market dynamics and emerging opportunities.

On the other hand, when a company embraces cloud services, it moves away from the traditional CapEx model as costs are restructured into an OpEx model. Instead of large upfront investments, organizations pay for cloud services on an ongoing, pay-as-you-use basis. This means they can scale their resources up or down as needed, without being tied to the constraints of owning assets. Cloud providers take on the responsibility of maintaining, upgrading, and securing the infrastructure, relieving organizations of those operational burdens. It allows greater agility and scalability. This flexibility enables companies to adjust their resources in near real time, whether they need to scale up during peak demand or scale

down during quieter periods. This dynamic allocation of resources supports rapid adaptation to evolving business requirements and ensures optimal cost management.

While the OpEx model offers several benefits, it also can present some challenges that organizations must consider. One significant challenge is the potential for overspending, particularly when there's a lack of control or accountability over expenses. Therefore, implementing robust controls and accountability measures is essential to ensure financial stability and efficiency.

Managing costs is fundamentally different in the cloud



Moving from CapEx to OpEx is a business-critical transformation that requires detailed planning and operational changes. To capitalize on the benefits of the cloud, organizations need to go through an organizational and cultural transformation that enables their teams to take advantage of the OpEx model. The FinOps framework provides an operating model to guide you on this journey.

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What is FinOps?

2 minutes

Overview

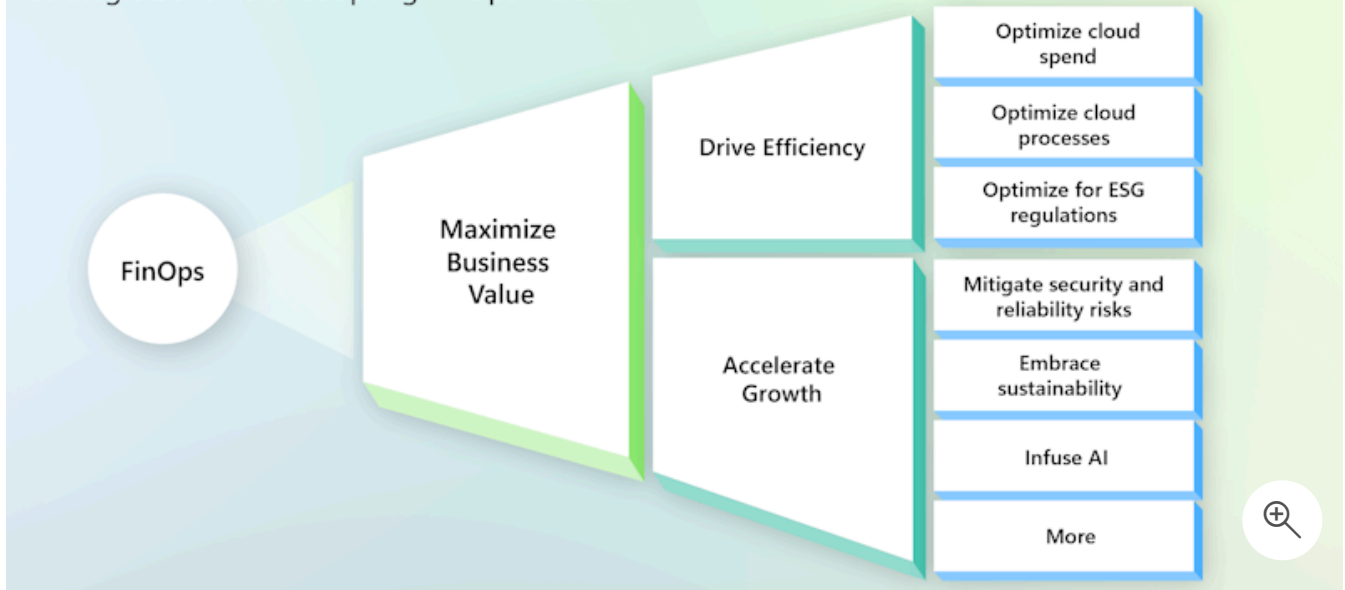
The FinOps Foundation defines FinOps as "an operational framework and cultural practice that maximizes the business value of cloud, enables timely data-driven decision making, and creates financial accountability through collaboration between engineering, finance, and business teams."

FinOps supports a cultural shift within organizations, fostering best practices aimed at enhancing efficiency and effectively managing cloud investments. It promotes accountability and data-driven decision-making to support business growth.

Contrary to popular belief, the primary objective of FinOps extends beyond mere cost optimization. It encompasses a comprehensive approach to managing cloud finances, focusing on optimizing cloud spending while also enhancing operational efficiency, resource utilization, and overall business agility. By embracing FinOps principles, organizations aim not only to control expenses, but also to align cloud investments with strategic goals, drive innovation, and foster continuous improvement in cloud management practices.

The primary objective of FinOps goes beyond cost reduction

Strategic benefits of adopting FinOps include:



FinOps Framework

The FinOps framework provides the operating model for establishing and excelling in the practice of FinOps. It helps organizations maximize the business value of cloud, enable timely data-driven decision making, and create financial accountability through collaboration between engineering, finance, and business teams. In fewer words, the framework is the operating model for FinOps.

You can think of the FinOps framework as the building blocks of a FinOps practice. In this metaphor, practitioners can choose and arrange their blocks in different ways, resulting in unique implementations of FinOps that are all constructed from the same components.

Like FinOps, the framework is evolving and informed by community experiences, contributions, and conversations. It's expounded by a growing library of stories, papers, and playbooks illustrating successful practices.

The primary components of the framework are:

- Principles
- Personas
- Phases
- Maturity model
- Domains
- Capabilities

For more information, see the [FinOps Framework](#) by the [FinOps Foundation](#).

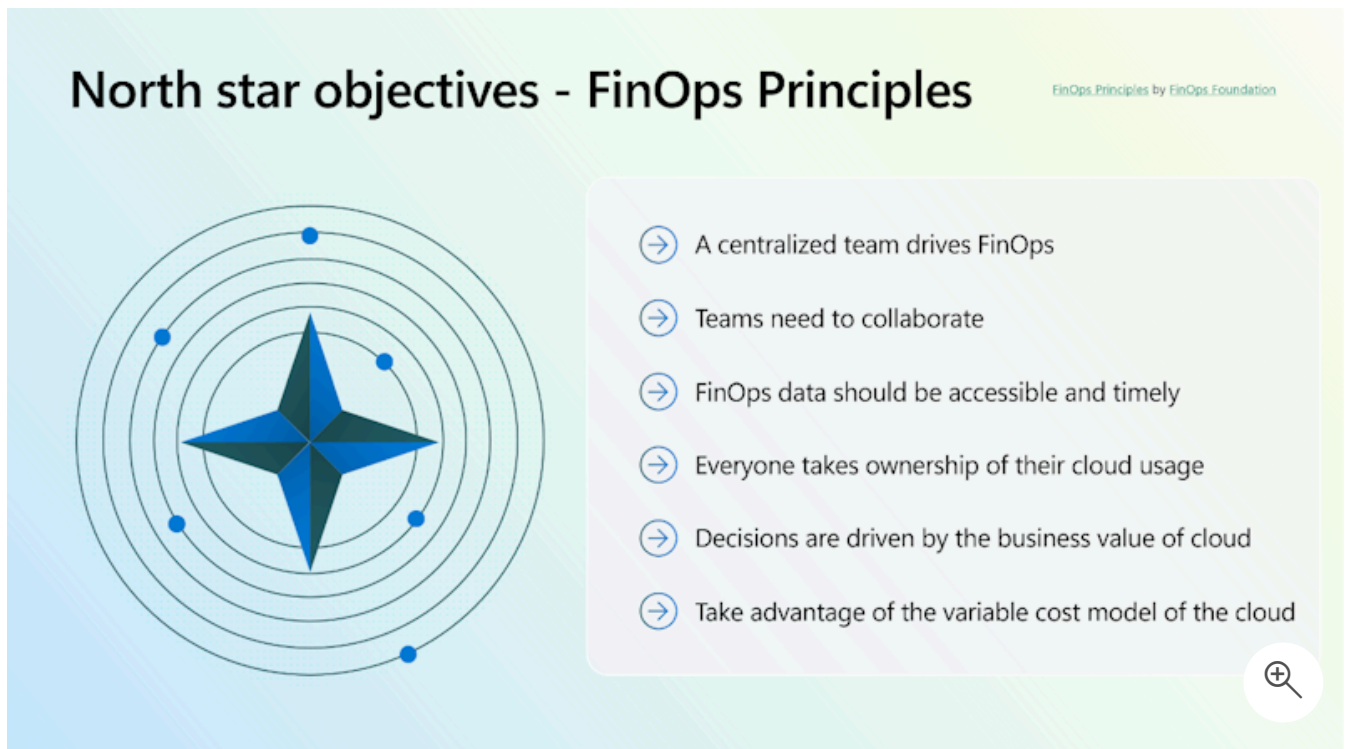
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FinOps Principles

3 minutes



The FinOps framework includes six principles that act as north star objectives of the FinOps practice. These principles are:

A centralized team drives FinOps

- The central team encourages, evangelizes, and enables best practices in a shared accountability model. In this regard, the FinOps central team is much like security, which has a central team, yet everyone remains responsible for their portion.
- Executive buy-in for FinOps and its practices and processes is required.
- Rate, commitment, and discount optimization are centralized to take advantage of economies of scale.
- Remove the need for engineers and operations teams to think about rate negotiations, allowing them to stay focused on usage optimization of their own environments.

Teams need to collaborate

- Finance, technology, product, and business teams work together in near real time as the cloud operates on a per-resource, per-second basis.
- Teams work together to continuously improve for efficiency and innovation.

FinOps data should be accessible and timely

- Process and share cost data as soon as it becomes available.
- Real-time visibility autonomously drives better cloud utilization.
- Fast feedback loops result in more efficient behavior.
- Consistent visibility into cloud spend is provided to all levels of the organization.
- Create, monitor, and improve real-time financial forecasting and planning.
- Trending and variance analysis helps explain why costs increased.
- Internal team benchmarking drives best practices and celebrates wins.
- Industry peer-level benchmarking assesses your company's performance.

Everyone takes ownership for their cloud usage

- Accountability of usage and cost is pushed to the edge, with engineers taking ownership of costs from architecture design to ongoing operations.
- Individual feature and product teams are empowered to manage their own cloud usage against their budget.
- Decentralize the decision making around cost-effective architecture, resource usage, and optimization.
- Technical teams must begin to consider cost as a new efficiency metric from the beginning of the software-development lifecycle.

Decisions are driven by business value of cloud

- Unit economic and value-based metrics demonstrate business impact better than aggregate spend.
- Make conscious trade-off decisions among cost, quality, and speed.
- Think of cloud as a driver of innovation.

Take advantage of the cloud's variable cost model

- You should view the cloud's variable cost model as an opportunity to deliver more value, not as a risk.
- Embrace just-in-time prediction, planning, and purchasing of capacity.
- Agile iterative planning is preferred over static long-term plans.
- Embrace proactive system design with continuous adjustments in cloud optimization over infrequent reactive cleanups.

For more information, see the [FinOps Principles](#) by the [FinOps Foundation](#) .

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FinOps Personas

2 minutes

To successfully adopt FinOps, you need to partner with different stakeholders in the organization. As part of this process, it's important to first identify the people who will contribute to the FinOps practice and understand their respective goals and motivations.

Core Personas

 Expand table

Persona	Primary Goal
FinOps Practitioner	Drive best practices into the organization through education, standardization, and cultural growth and support
Leadership	Assurance that cloud investments are aligned with business objectives
Product	Quickly bring new products and features to market with an accurate price point
Engineering	Deliver faster and high-quality services to the organization while maintaining business as usual
Finance	Accurately budget, forecast, and report cloud costs
Procurement	Cloud platform relationship management

Allied Personas

 Expand table

Persona	Overview
ITSM / ITIL	IT Service Management (ITSM) and Information Technology Infrastructure Library (ITIL) are responsible for collaborating with FinOps Practitioners to standardize and streamline IT service operations, improve service quality and reliability, and ensure that IT services meet agreed-upon service levels and that performance targets are balanced against cloud cost management priorities.
ITAM	IT Asset Management (ITAM) collaborates with FinOps Practitioners to achieve efficiency, transparency, and value in managing IT assets that impact cloud use by leveraging the expertise and data from both disciplines to optimize costs, ensure compliance, and support strategic business objectives balanced against cloud cost management priorities.
Sustainability	Sustainability collaborates with FinOps Practitioners to ensure cloud use optimizes for environmental impact, drive accountability, and accelerate progress towards broader sustainability goals in a way that is balanced against cloud cost management priorities.
Security	IT Security collaborates with FinOps Practitioners to leverage their FinOps expertise and insights to optimize cloud security spending, improve IT Security financial governance, and strengthen the organization's overall cloud security posture.
ITFM / TBM	IT Financial Management (ITFM) and Technology Business Management (TBM) are responsible for collaborating with FinOps Practitioners to provide transparency into IT spending and enable informed decision-making, ensuring cloud and traditional IT investments are aligned with business priorities, are cost-effective, and deliver measurable value to the organization.

For more information, see the [FinOps Personas](#) by the [FinOps Foundation](#).

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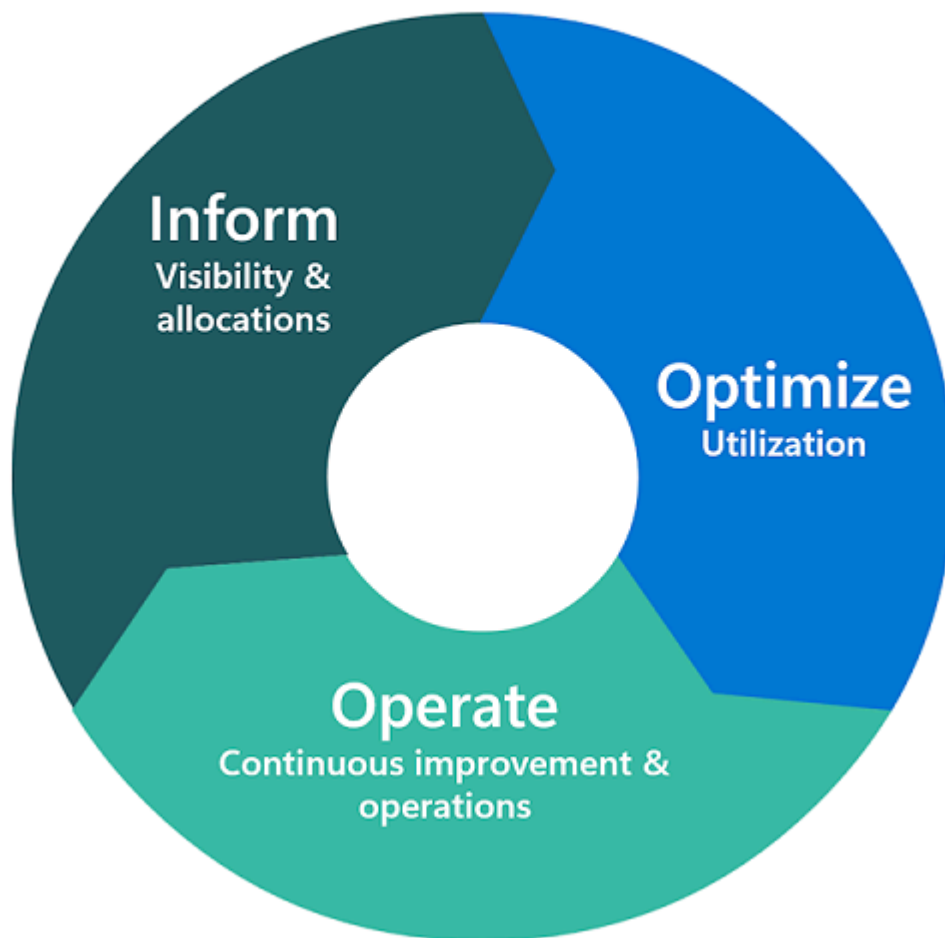
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FinOps Phases

2 minutes

The FinOps phases are designed as an iterative loop to support the customer journey in adopting and improving practices continuously. It consists of three phases—Inform, Optimize, and Operate—which don't need to follow a strict sequence. This flexibility allows organizations to customize it for their specific needs. Additionally, it's important to note that different teams might be in different stages based on various factors and objectives.

FinOps Phases



Inform

Visibility & Allocation

In the *Inform* phase, FinOps activities involve identifying data sources for cloud cost, usage, and efficiency data. Using this data for allocation, analysis and reporting empowers teams to develop capabilities in budgeting, forecast trends, build KPIs for benchmarking, and develop metrics that reveal the business value of an organization's cloud spend.

Optimize

Rates & Usage

In the *Optimize* phase, FinOps activities involve identifying opportunities to improve cloud efficiency using the data and capabilities developed in the Inform Phase.

Operate

Continuous Improvement & Usage

In the *Operate* phase, FinOps activities involve implementing organizational changes to operationalize FinOps using the data and capabilities developed in the Inform and Optimize phase. This includes establishing cloud governance policies, compliance monitoring, and empowering individuals through the development of training programs, team guidelines, and automation policies that are aligned with organizational objectives.

For more information, see the [FinOps Phases](#) by the [FinOps Foundation](#) .

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FinOps Maturity Model

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The FinOps Maturity Model serves as guidance for evaluating and steering an organization's progress in FinOps capabilities. It presents a structured method to comprehend and enhance cloud-management practices. The model delineates various stages, each signifying increasing levels of maturity and proficiency in cloud management. Let's delve into the stages within the FinOps Maturity Model:

Crawl

Characteristics:

- Very little reporting and tooling
- Measurements only provide insight into the benefits of maturing the capability
- Basic KPIs set for the measurement of success
- Basic processes and policies are defined around the capability
- Capability is understood but not followed by all the major teams within the organization
- Plans to address "low hanging fruit"

Walk

Characteristics:

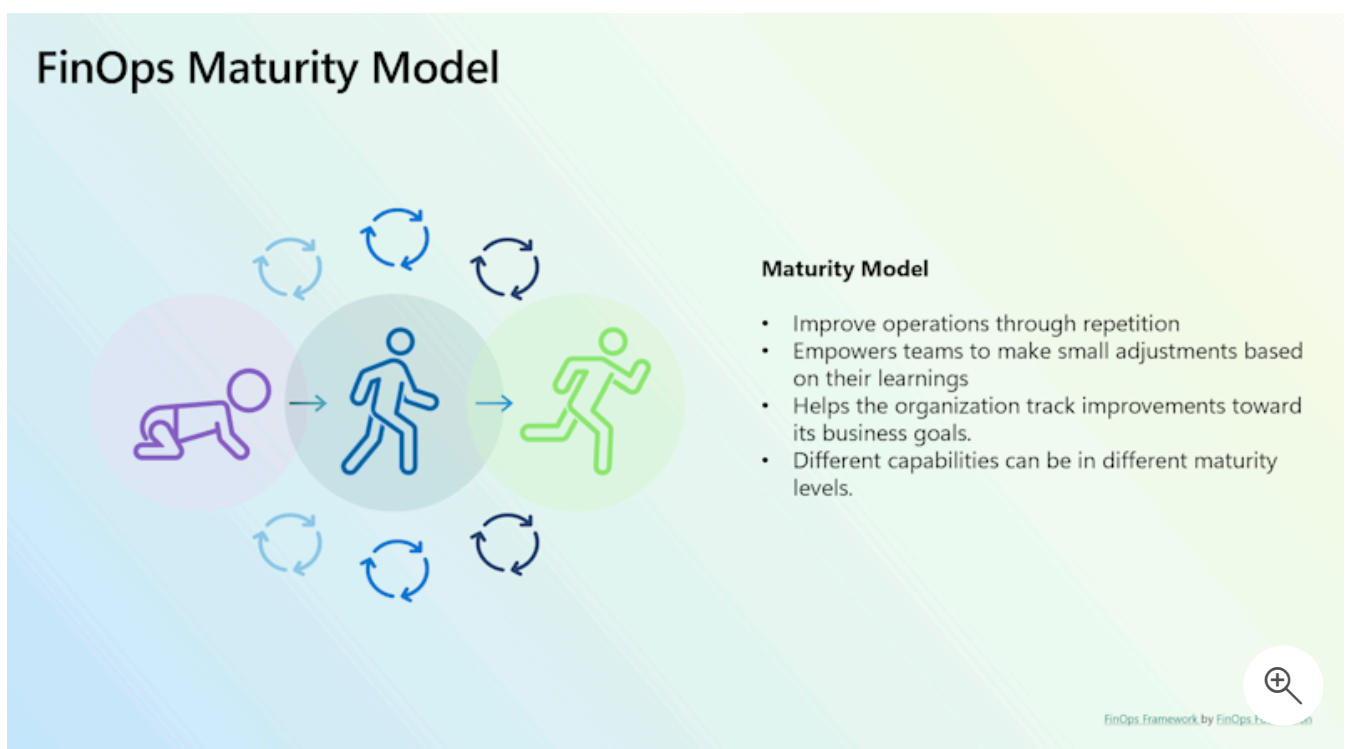
- Capability is understood and followed within the organization
- Difficult edge cases are identified, but decision to not address them is adopted
- Automation and/or processes cover most of the Capability requirements
- Most difficult edge cases (ones that threaten the financial well-being of the organization) are identified and effort to resolve has been estimated
- Medium to high goals/KPIs set on the measurement of success

Run

Characteristics:

- Capability is understood and followed by all teams within the organization
- Difficult edge cases are addressed
- Very high goals/KPIs set on the measurement of success
- Automation is the preferred approach

The FinOps Maturity Model offers organizations a roadmap to progress from basic cloud management to highly sophisticated, efficient practices. It aids in assessing their current state and setting goals for improvement. However, it's important to note that the goal of FinOps isn't to mandate every organization to reach the "Run" stage. It varies based on each organization's objectives and the business value it seeks to derive.



For more information, see the [FinOps Framework](#) by the [FinOps Foundation](#).

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FinOps Domains and Capabilities

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FinOps domains represent spheres of activity or knowledge. Every organization adopting FinOps engages in activities across all FinOps Domains. Each FinOps Domain comprises FinOps Capabilities, which outline functional activities that can be performed within that Domain.

The following sections group the FinOps domains and their respective capabilities.

Domain: Understand cloud usage and cost

Capabilities:

- Data ingestion
- Allocation
- Reporting and analytics
- Anomaly management

Domain: Quantify business value

Capabilities:

- Planning and estimating
- Forecasting
- Budgeting
- Benchmarking
- Unit economics

Domain: Optimize cloud usage and cost

Capabilities:

- Rate optimization

- Licensing and SaaS
- Workload optimization
- Architecting for cloud
- Cloud sustainability

Domain: Manage the FinOps practice

Capabilities:

- FinOps practice operations
- FinOps tooling and services
- FinOps assessment
- FinOps education and enablement
- Invoicing and chargeback
- Onboarding workloads
- Cloud policy and governance
- Intersecting disciplines

For more information, see [FinOps Domains](#) and [FinOps Capabilities](#) by the [FinOps Foundation](#) .

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FinOps Framework

Learn about what the FinOps Framework is and how you can use it to accelerate your cost management and optimization goals.

The [FinOps Framework](#) by the FinOps Foundation is a comprehensive set of best practices and principles. It provides a structured approach to implement a FinOps culture to:

- Help organizations manage their cloud costs more effectively
- Align cloud spending with business goals
- Drive greater business value from their cloud infrastructure

Microsoft's guidance is largely based on the FinOps Framework with a few enhancements based on the lessons learned from our vast ecosystem of Microsoft Cloud customers and partners. These extensions map cleanly back to FinOps Framework concepts and are intended to provide more targeted, actionable guidance for Microsoft Cloud customers and partners. We're working with the FinOps Foundation to incorporate our collective learnings back into the FinOps Framework.

In the next few sections, we cover the basic concepts of the FinOps Framework:

- The **principles** that should guide your FinOps efforts.
- The **scopes** to which you apply FinOps concepts.
- The **stakeholders** that should be involved.
- The **lifecycle** that you iterate through.
- The **capabilities** that you implement with stakeholders throughout the lifecycle.
- The **maturity model** that you use to measure growth over time.

Principles

Before digging into FinOps, it's important to understand the core principles that should guide your FinOps efforts. The FinOps community developed the principles by applying their collective experience to help you create a culture of shared accountability and transparency.

- **Teams need to collaborate** – Build a common focus on cost efficiency, processes, and cost decisions across teams that might not typically work closely together.

- **Business value drives technology decisions** – Balance cost decisions with business benefits including quality, speed, and business capability.
- **Everyone takes ownership for their technology usage** – Decentralize decisions about cloud resource usage and optimization, and drive technical teams to consider cost as well as uptime and performance.
- **FinOps data should be accessible, timely, and accurate** – Provide clear usage and cost data quickly, to the right people, to enable prompt decisions and forecasting.
- **FinOps should be enabled centrally** – Centralize management of FinOps practices for consistency, automation, and rate negotiations.
- **Take advantage of the variable cost model of the cloud** – Make continuous small adjustments in cloud usage and optimization.

For more information about FinOps principles, including tips from the experts, see [FinOps with Azure – Bringing FinOps to life through organizational and cultural alignment](#) .

Scopes

FinOps scopes conceptually represent the boundaries for which you apply your FinOps efforts. Your scope indicates the usage, spend, and sustainability management area of responsibility your team covers. This area of responsibility may also identify which personas are involved and what capabilities apply. Your team's scope is generally defined by and aligned with your organization's strategic goals and operational needs.

The scope or area of responsibility of your FinOps efforts may cover multiple technologies and platforms. Examples of things that may be part of your scope include, but are not limited to:

- Public, private, and hybrid clouds.
- Data centers.
- Cloud-based services offered by third-party providers (often referred to as "SaaS providers").
- Professional services, like consulting and training.

Your scope may also be a subset of these, if you focus solely on a single technology or even service. For instance, a team that only manages SQL databases may have a smaller area of responsibility than one that manages networking infrastructure for a global, multi-national company.

Ultimately, scopes represent what your team is responsible for managing or supporting, which impacts:

- What you need to know.
- Standards and processes you may need to apply to establish consistent execution and governance.
- Metrics and key performance indicators (KPIs) you need to measure success.
- What level of adoption you should target and expect within your organization.
- If and how you might automate tasks and processes to streamline your goals.

Your scope is determined based on your organization's business and technology needs and strategy to facilitate efficient execution of your FinOps practice to achieve your FinOps goals.

Stakeholders

FinOps requires a holistic and cross-functional approach that involves various stakeholders (or personas). They have different roles, responsibilities, and perspectives that influence how they use and optimize cloud resources and costs. Familiarize yourself with each role and identify the stakeholders within your organization. An effective FinOps program requires collaboration across all stakeholders:

- Core stakeholders:
 - **Procurement** – Source and purchase necessary resources, negotiate contracts, and managing vendor relationships.
 - **Finance** – Accurately budget, forecast, and report on cloud costs.
 - **Leadership** – Apply the strengths of the cloud to maximize business value.
 - **Business owners** – Drives strategic decision-making, budgeting, and understanding the financial impact of operational decisions.
 - **Product owners** – Define and prioritize the product backlog, aligning it with user needs and business value.
 - **Engineering teams** – Deliver high quality, cost-effective services.
 - **FinOps practitioners** – Educate, standardize, and promote FinOps best practices.
- Allied stakeholders:
 - **Sustainability practitioners** – Manage and reduce the environmental impact of the cloud resources.
 - **ITFM/TBM teams** (Information Technology Financial Management / Technology Business Management) – Cost management beyond cloud infrastructure and services.

- **ITSM/ITIL teams** (Information Technology Service Management / Information Technology Infrastructure Library) – Align IT services with business needs and deliver/support IT services that meet business goals.
- **ITAM teams** (Information Technology Asset Management) – Manage and optimize the purchase, deployment, maintenance, utilization, and disposal of software and hardware assets.
- **Security teams** – Ensure cloud operations and systems adhere to organizational security standards and policies.

Lifecycle

FinOps is an iterative, hierarchical process. Every team iterates through the FinOps lifecycle at their own pace, partnering with teams mentioned throughout all areas of the organization.

The FinOps Framework defines a simple lifecycle with three phases:

- **Inform** – Deliver cost visibility and create shared accountability through allocation, benchmarking, budgeting, and forecasting.
- **Optimize** – Reduce cloud waste and improve cloud efficiency by implementing various optimization strategies.
- **Operate** – Define, track, and monitor key performance indicators and governance policies that align cloud and business objectives.

Capabilities

The FinOps Framework includes capabilities that cover everything you need to perform FinOps tasks and manage a FinOps practice. Capabilities are organized into a set of related domains based on the goals of the capabilities. Each capability defines a functional area of activity and a set of tasks to support your FinOps practice.

- Understand usage and cost
 - [Data ingestion](#)
 - [Allocation](#)
 - [Reporting and analytics](#)
 - [Anomaly management](#)

- Quantify business value
 - [Planning and estimating](#)
 - [Forecasting](#)
 - [Budgeting](#)
 - [Benchmarking](#)
 - [Unit economics](#)
- Optimize usage and cost
 - [Architecting for the cloud](#)
 - [Workload optimization](#)
 - [Rate optimization](#)
 - [Licensing and SaaS](#)
 - [Cloud sustainability](#)
- Manage the FinOps practice
 - [FinOps education and enablement](#)
 - [FinOps practice operations](#)
 - [Onboarding workloads](#)
 - [Policy and governance](#)
 - [Invoicing and chargeback](#)
 - [FinOps assessment](#)
 - [FinOps tools and services](#)
 - [Intersecting frameworks](#)

Maturity model

As teams progress through the FinOps lifecycle, they naturally learn and grow, developing more mature practices with each iteration. Like the FinOps lifecycle, each team is at different levels of maturity based on their experience and focus areas.

The FinOps Framework defines a simple Crawl-Walk-Run maturity model, but the truth is that maturity is more complex and nuanced. Instead of focusing on a global maturity level, we believe it's more important to identify and assess progress against your goals in each area. At a high level, you will:

1. Identify the most critical capabilities for your business.

2. Define how important it is that each team has knowledge, process, success metrics, organizational alignment, and automation for each of the identified capabilities.
3. Evaluate each team's current knowledge, process, success metrics, organizational alignment, and level of automation based on the defined targets.
4. Identify steps that each team could take to improve maturity for each capability.
5. Set up regular check-ins to monitor progress and reevaluate the maturity assessment every 3-6 months.

Learn more at the FinOps Foundation

FinOps Foundation offers many resources to help you learn and implement FinOps. Join the FinOps community, explore training and certification programs, participate in community working groups, and more. For more information about FinOps, including useful playbooks, see the [FinOps Framework documentation](#) .

Give feedback

Let us know how we're doing with a quick review. We use these reviews to improve and expand FinOps tools and resources.

[Give feedback](#)

If you're looking for something specific, vote for an existing or create a new idea. Share ideas with others to get more votes. We focus on ideas with the most votes.

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Related content

Start your first or plan your next iteration:

- [Conduct a FinOps iteration](#)
-

Summary

2 minutes

FinOps is an operational framework and cultural practice that maximizes the business value of cloud, enables timely data-driven decision making, and creates financial accountability through collaboration between engineering, finance, and business teams. In the rapidly evolving digital landscape, cloud services have become integral to business operations, but they can also be a significant source of expenses. FinOps serves as a structured approach to tackle this challenge by promoting an organizational and cultural shift that empowers organizations to derive maximum value from their cloud investments.

At its core, FinOps revolves around six principles, and empowers individuals and teams to take ownership of their cloud spending while fostering a culture of cross-functional cooperation. Key components of the FinOps framework include its principals, phases, maturity model, personas, domains, and capabilities.

Learn more

Consider the following Microsoft resources to initiate or advance your FinOps journey:

- [FinOps Review Assessment](#)
Use FinOps guidance to assess your organization's capability gaps. Get recommendations for maximizing cloud business value using FinOps best practices.
- [FinOps Interactive Guides](#)
Gain hands-on experience with Microsoft solutions that enable various FinOps capabilities and best practices.
- [FinOps e-book](#)
This e-book guides you through the FinOps principles to highlight Microsoft solutions that can support your cloud journey.

- **FinOps Docs**

Find technical documentation and guidance to help make informed decisions and follow FinOps best practices.

- **FinOps on Azure**

Explore the range of Microsoft products and services designed to bolster your efforts in implementing FinOps best practices.

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